

NCR SHS ENROLLMENT FORECAST REPORT

Senior High School Enrollment Analysis & Resource Planning

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Forecast Period:	5 Years
Confidence Level:	95%
Model Type:	Facebook Prophet Time Series

EXECUTIVE SUMMARY

This comprehensive report analyzes three-year enrollment projections for NCR Senior High School. Based on historical enrollment data and advanced forecasting methodologies, the institution is expected to accommodate **106,715 total enrollees** by 2030, representing **17626 increase** (+19.8%) from the forecast period's initial year.

Key Projection: The analysis indicates need for **1385 academic classrooms**, **750 technical-vocational classrooms**, and a combined faculty of **1423 teachers** to maintain optimal learning conditions with a standard class size of 40 students per room.

KEY PERFORMANCE INDICATORS

Metric	Initial Year	Final Year (2030)	Change	Variance
Total Enrollees	89,089	106,715	+17,626	+19.8%
Academic Classrooms	1156	1385	+229	+19.8%
TVL Classrooms	626	750	+124	+19.8%
Total Faculty Required	1188	1423	+235	+19.8%

DETAILED 3-YEAR FORECAST

The following table presents the year-by-year enrollment projections with corresponding resource allocation requirements. Confidence intervals (2.5th-97.5th percentile) are provided to indicate forecast uncertainty ranges.

Year	Projected Enrollees	Confidence Interval	Academic Rooms	TVL Rooms	Total Faculty
2026	89,089	(79,230 - 98,915)	1156	626	1188
2027	93,493	(83,591 - 103,494)	1213	657	1247
2028	97,896	(87,897 - 107,716)	1270	688	1306
2029	102,312	(92,396 - 112,263)	1327	719	1364
2030	106,715	(96,817 - 116,617)	1385	750	1423

METHODOLOGY & TECHNICAL DETAILS

Forecasting Approach

This report utilizes **Facebook's Prophet**, an industry-standard time-series forecasting library designed for business metrics with clear seasonal patterns and trend changes. The model was trained on historical enrollment data and generates probabilistic forecasts with quantified uncertainty intervals, providing decision-makers with confidence bounds around point estimates.

Model Configuration:

- Interval Width: 95% (produces 2.5th to 97.5th percentile confidence bands)
- Seasonality Components: Disabled (sparse annual data precludes meaningful seasonal detection)
- Growth Model: Linear trend with automatic change-point detection
- Forecast Horizon: 3 years beyond historical data

Resource Allocation Calculations

Resource requirements are derived from enrollment projections using standardized educational metrics:

Classroom Distribution:

- Academic Stream: 30.0% of total enrollees / 50 students per room = Academic Classrooms Required
- Technical-Vocational Stream: 70.0% of total enrollees / 50 students per room = TVL Classrooms Required

Faculty Requirements (DepEd SHS Standard):

- 1 teacher handles 1.5 sections (DepEd SHS class programming guideline)
- Academic Teachers = Academic Rooms ÷ 1.5
- TVL Instructors = TVL Rooms ÷ 1.5

Key Assumptions:

- Standard classroom capacity: 50 students (aligned with DepEd standards)
- Academic/TVL enrollment ratio: 30.0%/70.0% (configurable in Admin Settings)
- No multi-shift or split-session scheduling
- Sections per teacher: 1.5 (DepEd SHS programming guideline)

Confidence Intervals & Uncertainty

The forecast confidence intervals reflect model uncertainty around point estimates. A 95% confidence interval indicates that there is a 95% probability the actual value will fall within the specified range, assuming the underlying patterns persist. This enables scenario planning:

Conservative Scenario: Use upper confidence bound for planning worst-case resource needs

Most Likely Scenario: Use point estimate for baseline planning

Optimistic Scenario: Use lower confidence bound for identifying cost-saving opportunities

STRATEGIC RECOMMENDATIONS

Immediate Actions (Next 6 Months)

1. **Infrastructure Planning:** Initiate procurement and construction planning for 229 additional academic rooms and 124 TVL facilities
2. **Faculty Recruitment:** Begin hiring processes for 235 additional educators with staggered onboarding
3. **Budget Allocation:** Prepare financial projections based on higher-bound confidence intervals to ensure sufficient resource allocation

Medium-Term Actions (6-24 Months)

1. **Classroom Development:** Establish infrastructure enhancement timeline aligned with enrollment growth projections
2. **Faculty Development:** Implement professional development programs and curriculum updates to support growing student population
3. **Capacity Monitoring:** Establish tracking mechanisms to compare actual vs. forecasted enrollments and adjust plans accordingly
4. **Resource Optimization:** Explore cost-effective solutions (shared facilities, optimal scheduling) within confidence bounds

Long-Term Strategy (2+ Years)

1. **Contingency Planning:** Develop alternative scenarios for upper and lower confidence bound outcomes
2. **Forecast Updates:** Refresh forecast quarterly with actual enrollment data to improve accuracy
3. **Quality Assurance:** Monitor student-teacher ratios and facility utilization to ensure educational quality is maintained
4. **Strategic Expansion:** Use multi-year trends to inform long-term campus expansion and program development plans

DISCLAIMER & NOTES

Forecast Accuracy: This forecast is based on historical enrollment patterns and assumes continuity of existing conditions. External factors (policy changes, demographic shifts, competing institutions) may impact actual results.

Data Quality: Forecast accuracy depends on the completeness and accuracy of input data. Missing or inconsistent data may affect results.

Review Frequency: Forecasts should be reviewed and updated quarterly with actual enrollment data to improve accuracy and identify emerging trends.

Model Limitations: Prophet is optimized for business metrics with clear trends and seasonal patterns. Unusual external shocks may not be captured by the model.

Professional Review: These forecasts should be reviewed by education administrators and policy makers before implementation. Actual resource planning should incorporate institutional priorities, budget constraints, and local factors.

This report was automatically generated on May 03, 2026 at 22:01:07. For inquiries or methodology clarifications, contact the forecasting team.